

## 1. General Description

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The EM74HC245; EM74HCT245 are 8-bit transceivers with 3-state outputs. The device features an output enable ( $\overline{OE}$ ) and send/receive (DIR) for direction control. A HIGH on  $\overline{OE}$  causes the outputs to assume a high-impedance OFF-state. Inputs include clamp diodes. This enables the use of current limiting resistors to interface inputs to voltages in excess of  $V_{CC}$ .

## 2. Features and Benefits

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- Wide supply voltage range from 2.0 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- Latch-up performance exceeds 250 mA
- Octal bidirectional bus interface
- Non-inverting 3-state outputs
- Complies with JEDEC standards:
  - JESD8C (2.7 V to 3.6 V)
  - JESD7A (2.0 V to 6.0 V)
- Input levels:
  - For EM74HC245: CMOS level
  - For EM74HCT245: TTL level
- ESD protection:
  - HBM ANSI/ESDA/JEDEC JS-001 Class 3A exceeds 5000 V
  - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

## EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state

### 3. Ordering Information

Table 1. Ordering information

| Type number  | Topside marking  | Package   |                                                                           |          |
|--------------|------------------|-----------|---------------------------------------------------------------------------|----------|
|              |                  | Name      | Description                                                               | Quantity |
| EM74HC245D   | HC245□<br>XYYWW  | SOP-20L   | plastic small outline package; 20 leads;<br>body width 7.5 mm             | 2000     |
| EM74HCT245D  | HCT245□<br>XYYWW |           |                                                                           |          |
| EM74HC245PW  | HC245□<br>XYYWW  | TSSOP-20L | plastic thin shrink small outline package; 20<br>leads; body width 4.4 mm | 3000     |
| EM74HCT245PW | HCT245□<br>XYYWW |           |                                                                           |          |

#### MARKING INFORMATION

**NOTE:** HC245□ = Serial number and Internal code; XYYWW = Vendor Code and Date Code.

HC245 □

Internal Code  
Serial number

X YY WW

Date Code -Week  
Date Code -Year  
Vendor Code

## 4. Function Diagram

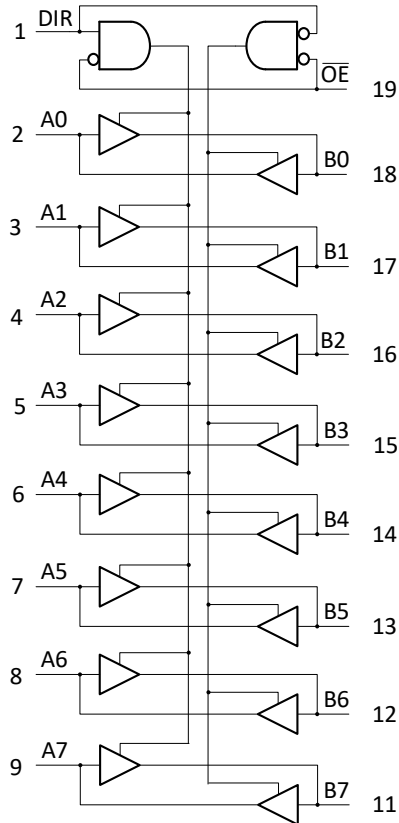


Fig. 1. Logic symbol

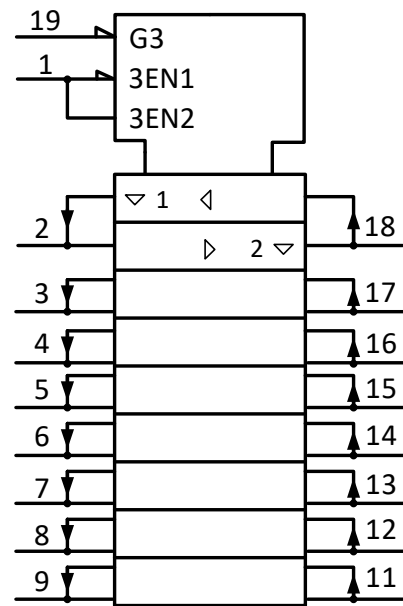
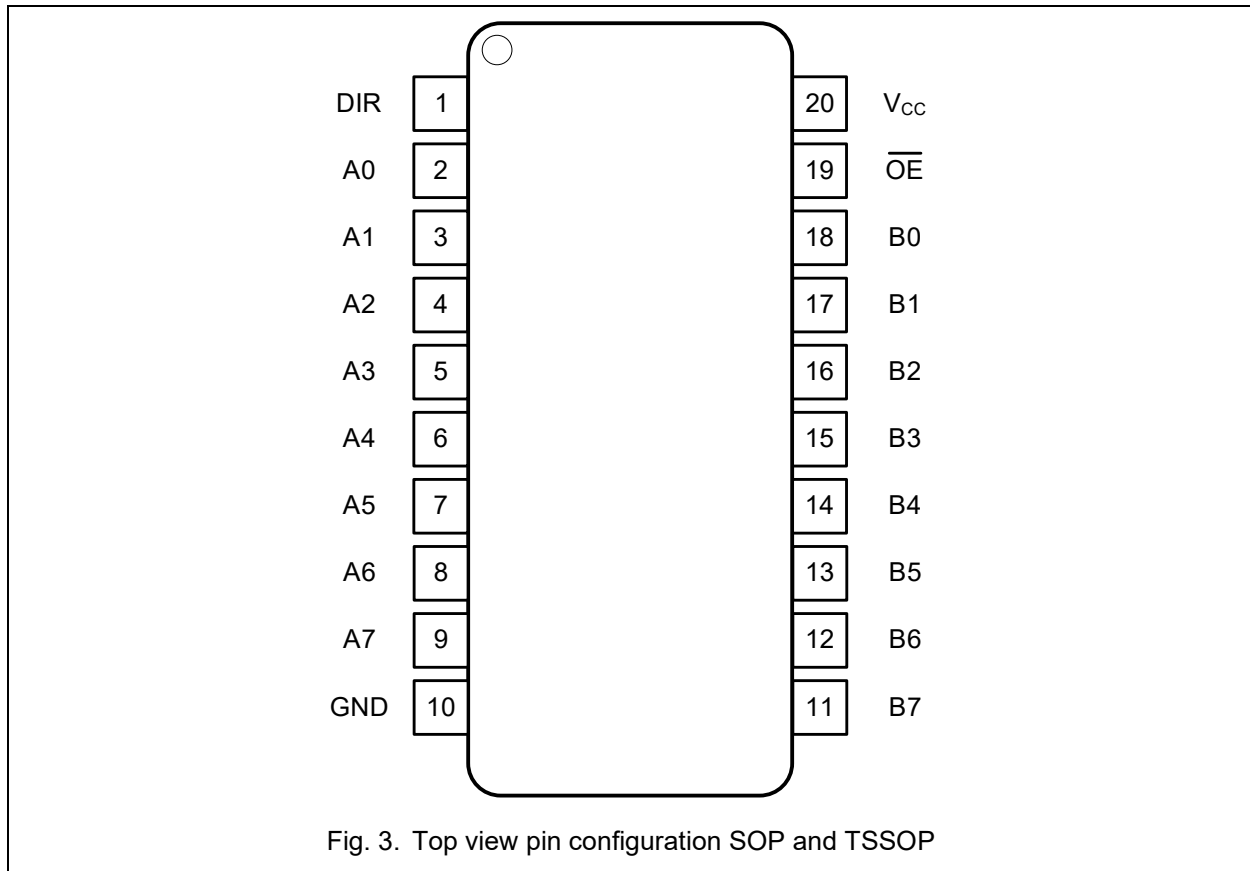


Fig. 2. IEC logic symbol

## 5. Pinning Information

### 5.1. Pinning



### 5.2. Pin description

**Table 2. Pin description**

| Symbol                         | Pin                            | Description                      |
|--------------------------------|--------------------------------|----------------------------------|
| DIR                            | 1                              | Direction control                |
| A0, A1, A2, A3, A4, A5, A6, A7 | 2, 3, 4, 5, 6, 7, 8, 9         | Data input/output                |
| GND                            | 10                             | Ground (0 V)                     |
| B7, B6, B5, B4, B3, B2, B1, B0 | 11, 12, 13, 14, 15, 16, 17, 18 | Data input/output                |
| $\overline{OE}$                | 19                             | Output enable input (active LOW) |
| V <sub>CC</sub>                | 20                             | Supply voltage                   |

## 6. Functional Description

**Table 3. Function table**

H = HIGH voltage level; L = LOW voltage level; X = don't care; Z = high-impedance OFF-state.

| Input           |     | Input/output |       |
|-----------------|-----|--------------|-------|
| $\overline{OE}$ | DIR | An           | Bn    |
| L               | L   | A = B        | input |
| L               | H   | input        | B = A |
| H               | X   | Z            | Z     |

## 7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol    | Parameter               | Conditions                                                 | Min  | Max      | Unit |
|-----------|-------------------------|------------------------------------------------------------|------|----------|------|
| $V_{CC}$  | supply voltage          |                                                            | -0.5 | 7.0      | V    |
| $I_{IK}$  | input clamping current  | $V_I < -0.5\text{ V}$ or $V_I > V_{CC} + 0.5\text{ V}$ [1] |      | $\pm 20$ | mA   |
| $I_{OK}$  | output clamping current | $V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$ [1] |      | $\pm 20$ | mA   |
| $I_O$     | output current          | $V_O = -0.5\text{ V}$ to $(V_{CC} + 0.5\text{ V})$         |      | $\pm 25$ | mA   |
| $I_{CC}$  | supply current          |                                                            |      | 50       | mA   |
| $I_{GND}$ | ground current          |                                                            | -50  |          | mA   |
| $P_{tot}$ | total power dissipation |                                                            |      | 500      | mW   |
| $T_{stg}$ | storage temperature     |                                                            | -65  | 150      | °C   |

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

## 8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

**Table 5. Recommended Operating Conditions**

Voltages are referenced to GND.

| Symbol              | Parameter                           | Conditions              | EM74HC245 |      |                 | EM74HCT245 |      |                 | Unit |
|---------------------|-------------------------------------|-------------------------|-----------|------|-----------------|------------|------|-----------------|------|
|                     |                                     |                         | Min       | Typ  | Max             | Min        | Typ  | Max             |      |
| V <sub>CC</sub>     | supply voltage                      |                         | 2.0       | 5.0  | 6.0             | 2.7        | 5.0  | 5.5             | V    |
| V <sub>I</sub>      | input voltage                       |                         | 0         |      | V <sub>CC</sub> | 0          |      | V <sub>CC</sub> | V    |
| V <sub>O</sub>      | output voltage                      |                         | 0         |      | V <sub>CC</sub> | 0          |      | V <sub>CC</sub> | V    |
| T <sub>amb</sub>    | ambient temperature                 |                         | -40       |      | 125             | -40        |      | 125             | °C   |
| $\Delta t/\Delta V$ | input transition rise and fall rate | V <sub>CC</sub> = 2.0 V |           |      | 625             |            |      |                 | ns/V |
|                     |                                     | V <sub>CC</sub> = 4.5 V |           | 1.67 | 139             |            | 1.67 | 139             | ns/V |
|                     |                                     | V <sub>CC</sub> = 6.0 V |           |      | 83              |            |      |                 | ns/V |

## EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state

# 9.Static Characteristics

**Table 6. Static characteristics**

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V). Typical values measured at  $T_{amb} = 25^{\circ}\text{C}$  (unless otherwise noted).

| Symbol           | Parameter                 | Conditions                                                                                                                 | -40 °C to +85 °C |     |      | -40 °C to +125 °C |      | Unit |
|------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------|-----|------|-------------------|------|------|
|                  |                           |                                                                                                                            | Min              | Typ | Max  | Min               | Max  |      |
| EM74HC245        |                           |                                                                                                                            |                  |     |      |                   |      |      |
| V <sub>IH</sub>  | HIGH-level input voltage  | V <sub>CC</sub> = 2.0 V                                                                                                    | 1.5              |     |      | 1.5               |      | V    |
|                  |                           | V <sub>CC</sub> = 4.5 V                                                                                                    | 3.15             |     |      | 3.15              |      | V    |
|                  |                           | V <sub>CC</sub> = 6.0 V                                                                                                    | 4.2              |     |      | 4.2               |      | V    |
| V <sub>IL</sub>  | LOW-level input voltage   | V <sub>CC</sub> = 2.0 V                                                                                                    |                  |     | 0.5  |                   | 0.5  | V    |
|                  |                           | V <sub>CC</sub> = 4.5 V                                                                                                    |                  |     | 1.35 |                   | 1.35 | V    |
|                  |                           | V <sub>CC</sub> = 6.0 V                                                                                                    |                  |     | 1.8  |                   | 1.8  | V    |
| V <sub>OH</sub>  | HIGH-level output voltage | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub>                                                                        |                  |     |      |                   |      |      |
|                  |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 2.0 V                                                                           | 1.9              |     |      | 1.9               |      | V    |
|                  |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 4.5 V                                                                           | 4.4              |     |      | 4.4               |      | V    |
|                  |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 6.0 V                                                                           | 5.9              |     |      | 5.9               |      | V    |
|                  |                           | I <sub>O</sub> = -6.0 mA; V <sub>CC</sub> = 4.5 V                                                                          | 3.84             |     |      | 3.7               |      | V    |
|                  |                           | I <sub>O</sub> = -7.8 mA; V <sub>CC</sub> = 6.0 V                                                                          | 5.34             |     |      | 5.2               |      | V    |
| V <sub>OL</sub>  | LOW-level output voltage  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub>                                                                        |                  |     |      |                   |      |      |
|                  |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 2.0 V                                                                            |                  |     | 0.1  |                   | 0.1  | V    |
|                  |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 4.5 V                                                                            |                  |     | 0.1  |                   | 0.1  | V    |
|                  |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 6.0 V                                                                            |                  |     | 0.1  |                   | 0.1  | V    |
|                  |                           | I <sub>O</sub> = 6.0 mA; V <sub>CC</sub> = 4.5 V                                                                           |                  |     | 0.33 |                   | 0.4  | V    |
|                  |                           | I <sub>O</sub> = 7.8 mA; V <sub>CC</sub> = 6.0 V                                                                           |                  |     | 0.33 |                   | 0.4  | V    |
| I <sub>I</sub>   | input leakage current     | V <sub>I</sub> = V <sub>CC</sub> or GND ;<br>V <sub>CC</sub> = 6.0 V                                                       |                  |     | ±1   |                   | ±1   | μA   |
| I <sub>OZ</sub>  | OFF-state output current  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub> ; V <sub>CC</sub> = 6.0 V ;<br>V <sub>O</sub> = V <sub>CC</sub> or GND |                  |     | ±5   |                   | ±10  | μA   |
| I <sub>CC</sub>  | supply current            | V <sub>I</sub> = V <sub>CC</sub> or GND ; I <sub>O</sub> = 0 A ;<br>V <sub>CC</sub> = 6.0 V                                |                  |     | 20   |                   | 40   | μA   |
| C <sub>I</sub>   | input capacitance         |                                                                                                                            |                  | 4.0 |      |                   |      | pF   |
| C <sub>I/O</sub> | Input/output capacitance  |                                                                                                                            |                  | 7.7 |      |                   |      | pF   |

## EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state

| Symbol           | Parameter                 | Conditions                                                                                                                                               | -40 °C to +85 °C |     |      | -40 °C to +125 °C |     | Unit |
|------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----|------|-------------------|-----|------|
|                  |                           |                                                                                                                                                          | Min              | Typ | Max  | Min               | Max |      |
| EM74HCT245       |                           |                                                                                                                                                          |                  |     |      |                   |     |      |
| V <sub>IH</sub>  | HIGH-level input voltage  | V <sub>CC</sub> = 3 V                                                                                                                                    | 2.0              |     |      | 2.0               |     | V    |
|                  |                           | V <sub>CC</sub> = 4.5 V                                                                                                                                  | 2.0              |     |      | 2.0               |     | V    |
|                  |                           | V <sub>CC</sub> = 5.5 V                                                                                                                                  | 2.1              |     |      | 2.1               |     | V    |
| V <sub>IL</sub>  | LOW-level input voltage   | V <sub>CC</sub> = 3 V                                                                                                                                    |                  |     | 0.8  |                   | 0.8 | V    |
|                  |                           | V <sub>CC</sub> = 4.5 V                                                                                                                                  |                  |     | 0.8  |                   | 0.8 | V    |
|                  |                           | V <sub>CC</sub> = 5.5 V                                                                                                                                  |                  |     | 0.8  |                   | 0.8 | V    |
| V <sub>OH</sub>  | HIGH-level output voltage | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub> ;                                                                                                    |                  |     |      |                   |     |      |
|                  |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 3 V                                                                                                           | 2.9              |     |      | 2.9               |     | V    |
|                  |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 4.5 V                                                                                                         | 4.4              |     |      | 4.4               |     | V    |
|                  |                           | I <sub>O</sub> = -6.0 mA; V <sub>CC</sub> = 4.5 V                                                                                                        | 3.84             |     |      | 3.7               |     | V    |
| V <sub>OL</sub>  | LOW-level output voltage  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub> ;                                                                                                    |                  |     |      |                   |     |      |
|                  |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 3 V                                                                                                            |                  |     | 0.1  |                   | 0.1 | V    |
|                  |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 4.5 V                                                                                                          |                  |     | 0.1  |                   | 0.1 | V    |
|                  |                           | I <sub>O</sub> = 6.0 mA; V <sub>CC</sub> = 4.5 V                                                                                                         |                  |     | 0.33 |                   | 0.4 | V    |
| I <sub>I</sub>   | input leakage current     | V <sub>I</sub> = V <sub>CC</sub> or GND ;<br>V <sub>CC</sub> = 5.5 V                                                                                     |                  |     | ±1   |                   | ±1  | μA   |
| I <sub>OZ</sub>  | OFF-state output current  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub> ; V <sub>CC</sub> = 5.5 V ;<br>V <sub>O</sub> = V <sub>CC</sub> or GND                               |                  |     | ±5   |                   | ±10 | μA   |
| I <sub>CC</sub>  | supply current            | V <sub>I</sub> = V <sub>CC</sub> or GND; I <sub>O</sub> = 0 A;<br>V <sub>CC</sub> = 5.5 V                                                                |                  |     | 20   |                   | 40  | μA   |
| ΔI <sub>CC</sub> | additional supply current | per pin ; V <sub>I</sub> = V <sub>CC</sub> - 2.1 V; I <sub>O</sub> = 0 A;<br>other inputs at V <sub>CC</sub> or GND;<br>V <sub>CC</sub> = 4.5 V to 5.5 V |                  |     | 400  |                   | 490 | μA   |
| C <sub>I</sub>   | input capacitance         |                                                                                                                                                          |                  | 4.0 |      |                   |     | pF   |
| C <sub>I/O</sub> | Input/output capacitance  |                                                                                                                                                          |                  | 7.7 |      |                   |     | pF   |

## EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state

# 10. Dynamic Characteristics

**Table 7. Dynamic characteristics**

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 6. Typical values measured at  $T_{amb} = 25^{\circ}\text{C}$  (unless otherwise noted).

| Symbol           | Parameter                     | Conditions                                              | -40 °C to +85 °C |     |     | -40 °C to +125 °C |     | Unit |
|------------------|-------------------------------|---------------------------------------------------------|------------------|-----|-----|-------------------|-----|------|
|                  |                               |                                                         | Min              | Typ | Max | Min               | Max |      |
| EM74HC245        |                               |                                                         |                  |     |     |                   |     |      |
| t <sub>pd</sub>  | propagation delay             | An to Bn or Bn to An; see Fig. 4 [1]                    |                  |     |     |                   |     |      |
|                  |                               | V <sub>CC</sub> = 2.0 V                                 |                  |     | 35  |                   | 40  | ns   |
|                  |                               | V <sub>CC</sub> = 4.5 V                                 |                  |     | 20  |                   | 25  | ns   |
|                  |                               | V <sub>CC</sub> = 6.0 V                                 |                  |     | 15  |                   | 20  | ns   |
| t <sub>en</sub>  | enable time                   | $\overline{OE}$ to An or Bn; see Fig. 5 [2]             |                  |     |     |                   |     |      |
|                  |                               | V <sub>CC</sub> = 2.0 V                                 |                  |     | 35  |                   | 40  | ns   |
|                  |                               | V <sub>CC</sub> = 4.5 V                                 |                  |     | 20  |                   | 25  | ns   |
|                  |                               | V <sub>CC</sub> = 6.0 V                                 |                  |     | 15  |                   | 20  | ns   |
| t <sub>dis</sub> | disable time                  | $\overline{OE}$ to An or Bn; see Fig. 5 [3]             |                  |     |     |                   |     |      |
|                  |                               | V <sub>CC</sub> = 2.0 V                                 |                  |     | 35  |                   | 40  | ns   |
|                  |                               | V <sub>CC</sub> = 4.5 V                                 |                  |     | 20  |                   | 25  | ns   |
|                  |                               | V <sub>CC</sub> = 6.0 V                                 |                  |     | 15  |                   | 20  | ns   |
| t <sub>t</sub>   | transition time               | see Fig. 4 [4]                                          |                  |     |     |                   |     |      |
|                  |                               | V <sub>CC</sub> = 2.0 V                                 |                  |     | 9   |                   | 11  | ns   |
|                  |                               | V <sub>CC</sub> = 4.5 V                                 |                  |     | 6   |                   | 8   | ns   |
|                  |                               | V <sub>CC</sub> = 6.0 V                                 |                  |     | 4   |                   | 5   | ns   |
| C <sub>PD</sub>  | power dissipation capacitance | per buffer; V <sub>I</sub> = GND to V <sub>CC</sub> [5] |                  | 18  |     |                   |     | pF   |

## EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state

| Symbol           | Parameter                     | Conditions                                                          | -40 °C to +85 °C |     |     | -40 °C to +125 °C |     | Unit |
|------------------|-------------------------------|---------------------------------------------------------------------|------------------|-----|-----|-------------------|-----|------|
|                  |                               |                                                                     | Min              | Typ | Max | Min               | Max |      |
| EM74HCT245       |                               |                                                                     |                  |     |     |                   |     |      |
| t <sub>pd</sub>  | propagation delay             | An to Bn or Bn to An; V <sub>CC</sub> = 4.5V; see Fig. 4 [1]        |                  |     | 20  |                   | 25  | ns   |
| t <sub>en</sub>  | enable time                   | $\overline{OE}$ to An or Bn; V <sub>CC</sub> = 4.5V; see Fig. 5 [2] |                  |     | 20  |                   | 25  | ns   |
| t <sub>dis</sub> | disable time                  | $\overline{OE}$ to An or Bn; V <sub>CC</sub> = 4.5V see Fig. 5 [3]  |                  |     | 20  |                   | 25  | ns   |
| t <sub>t</sub>   | transition time               | V <sub>CC</sub> =4.5V; see Fig. 4; [4]                              |                  |     | 6   |                   | 8   | ns   |
| C <sub>PD</sub>  | power dissipation capacitance | per buffer; V <sub>I</sub> = GND to V <sub>CC</sub> – 1.5 V [5]     |                  | 18  |     |                   |     | pF   |

[1]  $t_{pd}$  is the same as  $t_{PHL}$  and  $t_{PLH}$ .

[2]  $t_{en}$  is the same as  $t_{PZH}$  and  $t_{PZL}$ .

[3]  $t_{dis}$  is the same as  $t_{PLZ}$  and  $t_{PHZ}$ .

[4]  $t_t$  is the same as  $t_{THL}$  and  $t_{TLH}$ .

[5]  $C_{PD}$  is used to determine the dynamic power dissipation ( $P_D$  in  $\mu W$ ).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o)$  where:

$f_i$  = input frequency in MHz;

$f_o$  = output frequency in MHz;

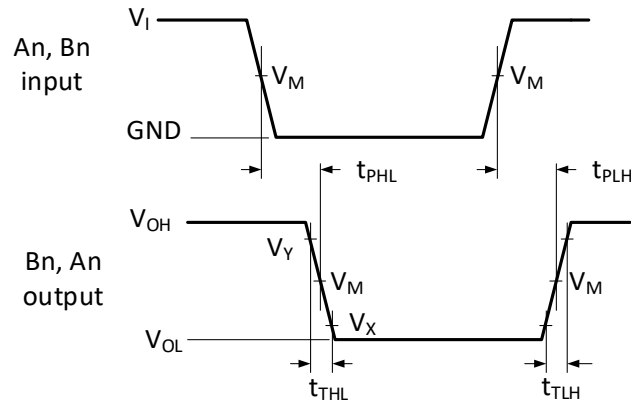
$C_L$  = output load capacitance in pF;

$V_{CC}$  = supply voltage in V;

$N$  = number of inputs switching;

$\Sigma(C_L \times V_{CC}^2 \times f_o)$  = sum of outputs.

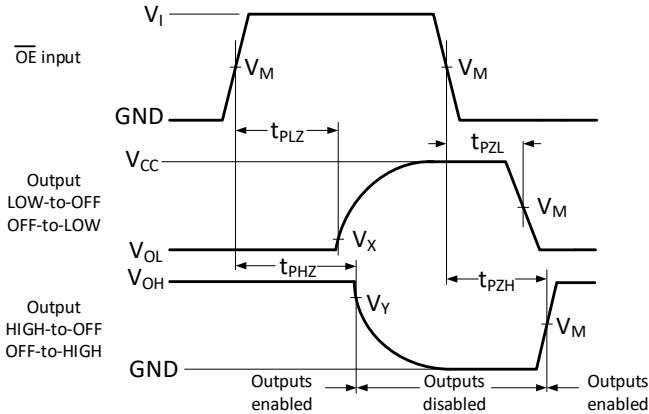
## 10.1. Waveforms and test circuit



Measurement points are given in Table 8.

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

Fig. 4. Input (An, Bn) to output (Bn, An) propagation delays and output transition times



Measurement points are given in Table 8.

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

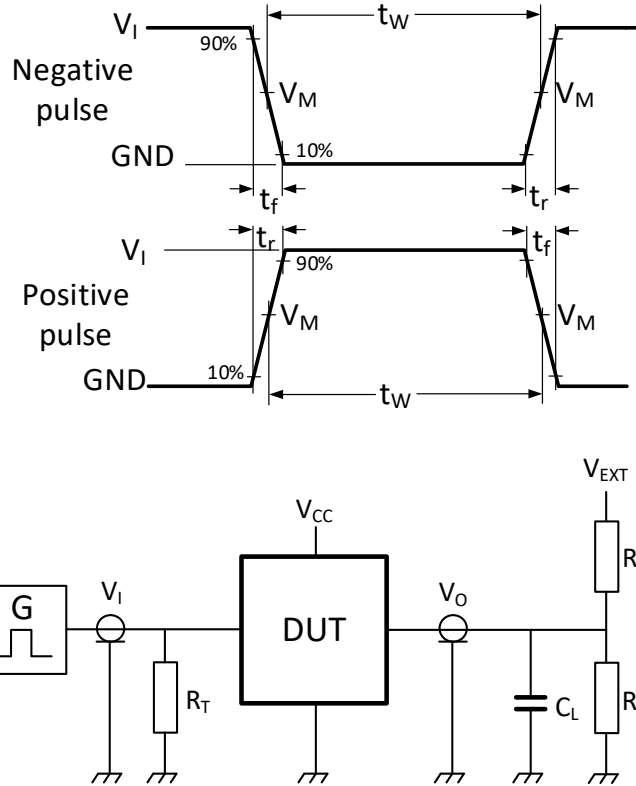
Fig. 5. 3-state enable and disable times

**Table 8. Measurement points**

| Type       | Input       | Output      |             |             |
|------------|-------------|-------------|-------------|-------------|
|            | $V_M$       | $V_M$       | $V_X$       | $V_Y$       |
| EM74HC245  | $0.5V_{CC}$ | $0.5V_{CC}$ | $0.1V_{CC}$ | $0.9V_{CC}$ |
| EM74HCT245 | 1.3 V       | 1.3 V       | $0.1V_{CC}$ | $0.9V_{CC}$ |

## EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state



Test data is given in Table 9.

Definitions for test circuit:

$R_L$  = Load resistance.

$C_L$  = Load capacitance including jig and probe capacitance.

$R_T$  = Termination resistance should be equal to the output impedance  $Z_o$  of the pulse generator.

$V_{EXT}$  = External voltage for measuring switching times.

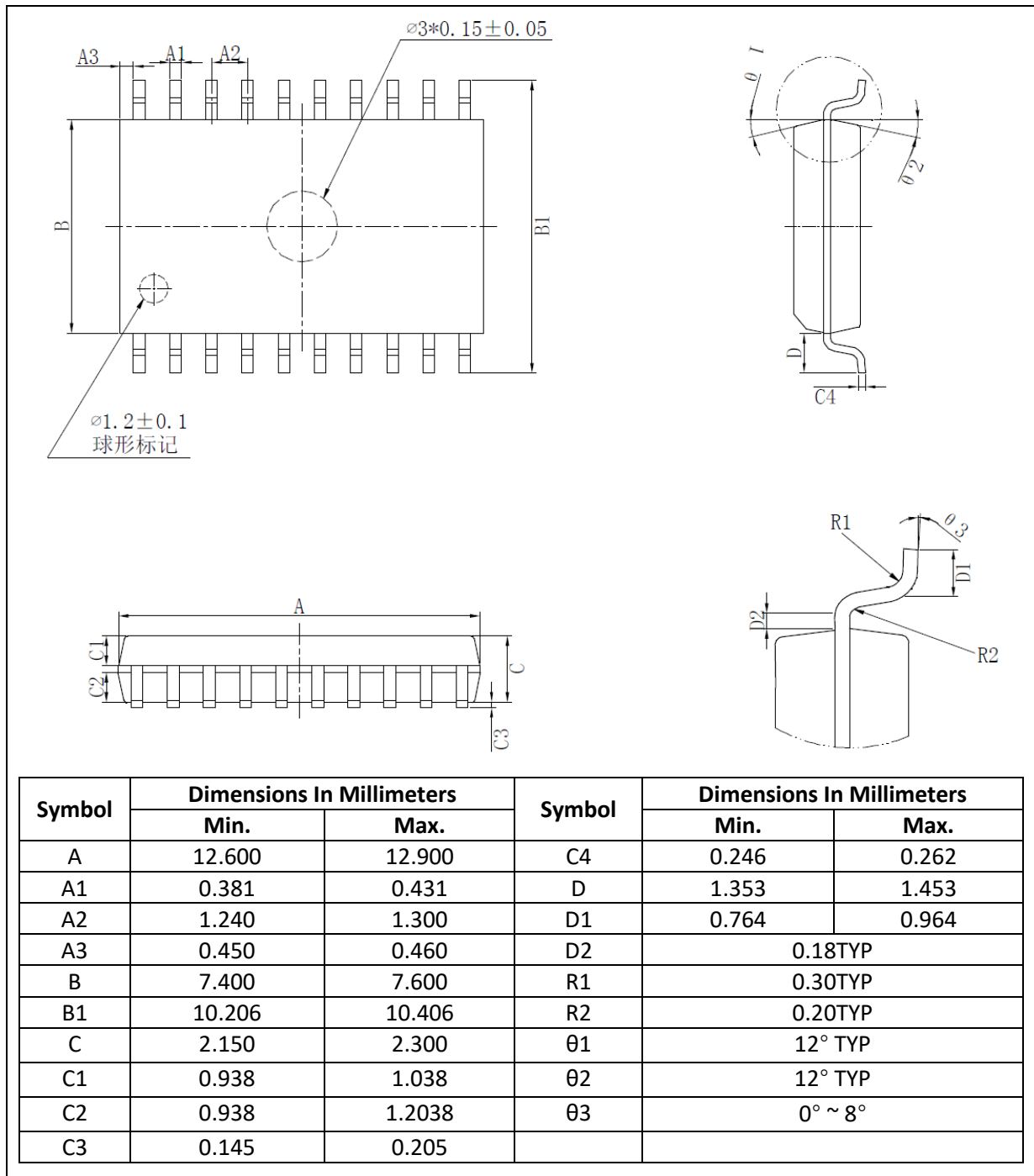
Fig. 6. Test circuit for measuring switching times

Table 9. Test data

| Type       | Input    |                       | Load  |              | $V_{EXT}$          |                    |                    |
|------------|----------|-----------------------|-------|--------------|--------------------|--------------------|--------------------|
|            | $V_I$    | $t_r = t_f$           | $C_L$ | $R_L$        | $t_{PLH}, t_{PHL}$ | $t_{PZH}, t_{PHZ}$ | $t_{PZL}, t_{PLZ}$ |
| EM74HC245  | $V_{CC}$ | $\leq 2.5 \text{ ns}$ | 50 pF | 500 $\Omega$ | open               | GND                | $2V_{CC}$          |
| EM74HCT245 | 3 V      | $\leq 2.5 \text{ ns}$ | 50 pF | 500 $\Omega$ | open               | GND                | $2V_{CC}$          |

# 11. Package Outline

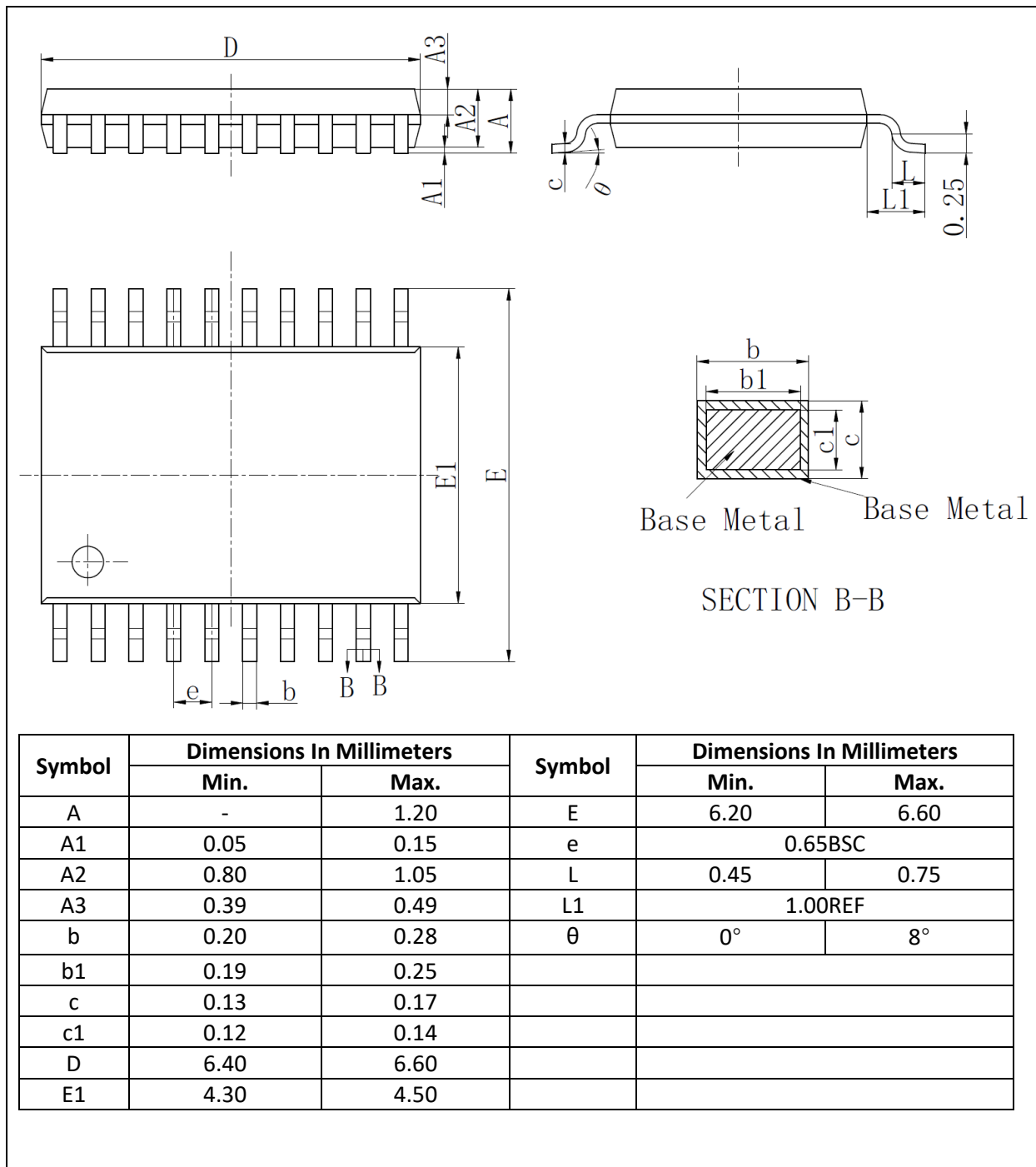
## SOP-20L



# EM74HC245; EM74HCT245

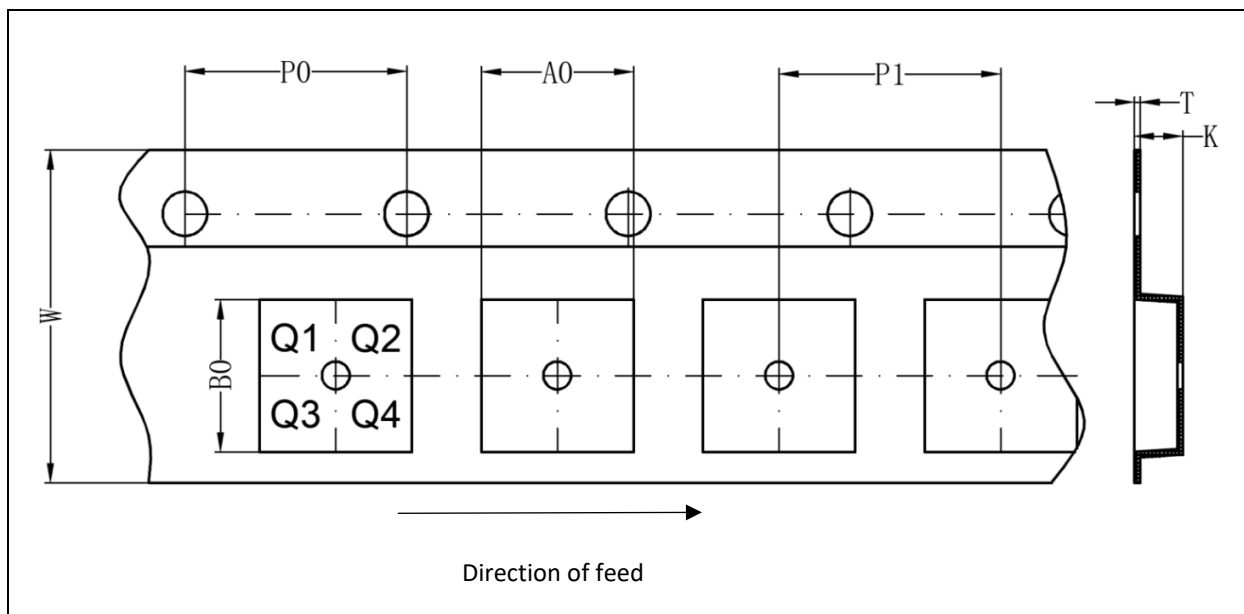
Octal buffer/line driver; 3-state

## TSSOP-20L



## 12. Tape and Reel Information

### 12.1. Carrier tape dimensions



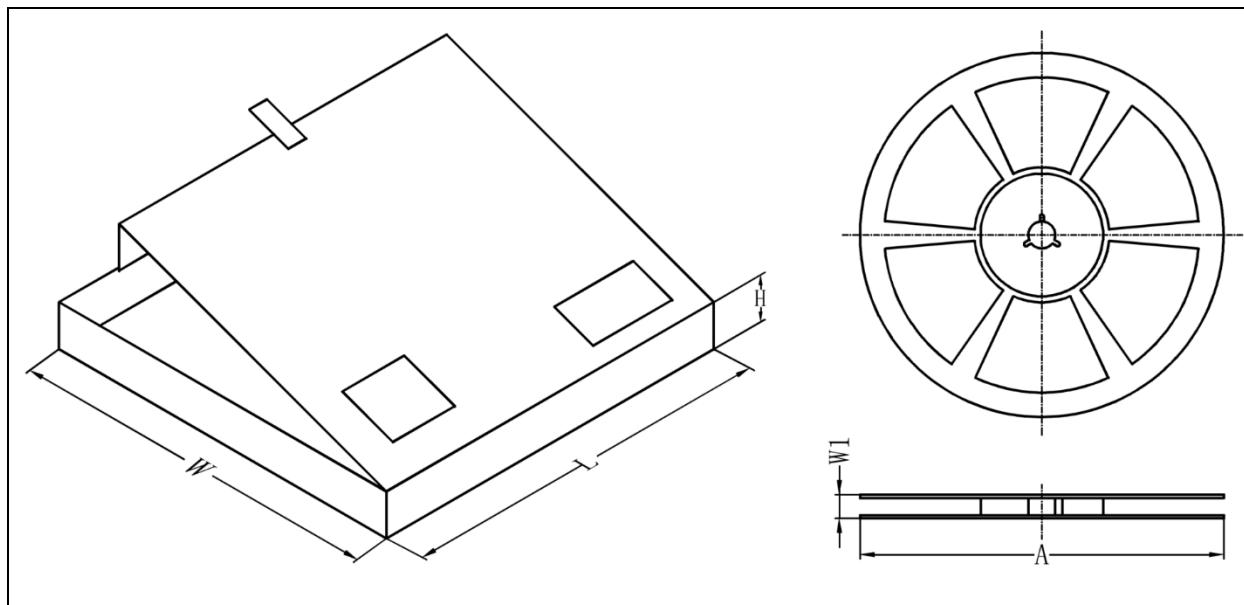
**Table 10. Carrier tape dimensions**

| Package version | A0(mm) | B0(mm) | K0(mm) | T(mm) | P1(mm) | W(mm) | P0(mm) | PIN 1 |
|-----------------|--------|--------|--------|-------|--------|-------|--------|-------|
| SOP-20L         | 10.65  | 13.2   | 3.2    | 0.3   | 12     | 24    | 4      | Q1    |
| TSSOP-20L       | 6.85   | 6.85   | 1.7    | 0.22  | 8      | 12    | 4      | Q1    |

## EM74HC245; EM74HCT245

Octal buffer/line driver; 3-state

### 12.2. Reel and box dimensions



**Table 11. Dimensions and quantities**

| Package version | Type NO. ending | Reel Dimension A (mm) | Reel Width W1 (mm) | SPQ (pcs) [1] | Reels per box | Outer box dimensions L×W×H(mm) [2] |
|-----------------|-----------------|-----------------------|--------------------|---------------|---------------|------------------------------------|
| SOP-20L         | D               | 330                   | 30.4               | 2000          | 1             | 358x340x50                         |
| TSSOP-20L       | PW              | 330                   | 18.4               | 3000          | 1             | 358x340x50                         |

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

## 13. Abbreviations

**Table 12. Abbreviations**

| Acronym | Description                             |
|---------|-----------------------------------------|
| CMOS    | Complementary Metal-Oxide Semiconductor |
| DUT     | Device Under Test                       |
| ESD     | ElectroStatic Discharge                 |
| HBM     | Human Body Model                        |
| CDM     | Charged Device Model                    |

## 14. Revision History

**Table 13. Revision history**

| Document ID            | Release Date                                                                                                                                      | Data sheet status | Change notice | Supersedes             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------|------------------------|
| EM74HC_HCT245 Rev. 1.1 | Dec 15, 2025                                                                                                                                      | Product datasheet |               | EM74HC_HCT245 Rev. 1.0 |
| Modifications:         | <ul style="list-style-type: none"><li>Table 1: topside marking information updated.</li><li>Section12: Added tape and reel information.</li></ul> |                   |               |                        |
| EM74HC_HCT245 Rev. 1.0 | Apr 18, 2025                                                                                                                                      | Product datasheet |               |                        |